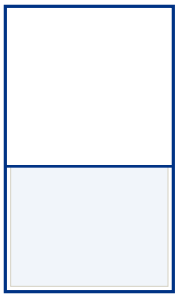
Density and measurement

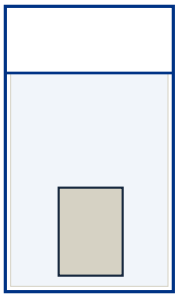
Use mass and volume together, with units and precision that match the measurements.

Mass: 54.0 g

Dry, insoluble solid; fully submerged; no trapped gas or liquid loss.



20.0 mL before



40.0 mL after

Volume = 40.0 - 20.0 = 20.0 mL

Density = mass / volume = 2.70 g/mL

Measurement idea	Consequence
Compare with water (1.00 g/mL)	A rigid nonreacting solid at 2.70 g/mL sinks, assuming no trapped gas.
Air bubbles cling to object	Apparent displaced volume is too large; calculated density is too low.
Use the final cylinder reading alone	That includes original water volume and overestimates the solid volume.

Units and significant figures

1 mL = 1 cm3. 1.25 g/mL = 1.25 x 10^3 kg/m3.
Subtract readings to the measured decimal place; divide using the appropriate significant figures. Matching density supports an identification but does not prove it. Cylinder sketches are not calibrated scale drawings.

Density and measurement

Self-check questions and explanations. Refer to the visual guide on the front.

What solid volume is indicated by 20.0 mL before and 40.0 mL after?

Volume = 40.0 mL - 20.0 mL = 20.0 mL, preserving the tenths place of the readings. Difference in liquid level gives displaced volume for a fully submerged insoluble object.

What density follows from a 54.0 g mass?

Density = mass/volume = 54.0 g / 20.0 mL = 2.70 g/mL (3 significant figures). Volume is the difference of readings, not the final reading alone.

Would the solid sink in water at 1.00 g/mL?

It sinks under the stated assumptions because 2.70 g/mL is greater than 1.00 g/mL. Density compares mass per equal volume, not mass alone.

How does 1.25 g/mL convert to kg/m3?

1.25 g/cm3 x (0.001 kg/g) x (1,000,000 cm3/m3) = 1.25 x 10^3 kg/m3. One cubic meter holds one million cubic centimeters while each gram is 0.001 kg.

A 2.00 x 3.00 x 4.00 cm block has mass 46.8 g: what is its density?

Volume = 24.0 cm3; density = 46.8/24.0 = 1.95 g/cm3. It is less dense than the first solid. Equal density alone is not conclusive identification because materials can share similar density and conditions matter.

How do clinging air bubbles affect calculated density?

Bubbles add displaced volume without adding the corresponding solid mass. Apparent volume is too large; mass divided by that overestimate gives a density that is too low.